

Document Intelligence Connector

Product Summary & Current Build Status

1. Product summary

Document Intelligence Connector is middleware between inbound business documents and the client's accounting system (**Zoho Books** today).

It is **not**:

- a bookkeeping app
- a Zoho competitor
- "just OCR"

Point of the product: cheap extraction + **judgment** (is this trustworthy?) + controlled ERP push + human review when unsure + full audit trail.

POC "done" looks like: an invoice becomes a correct Zoho bill, with a clear record of what was extracted, what passed/failed, and who approved it — without manual re-keying.

2. How it's supposed to work (end vision)

1. Documents arrive (Email / Drive / Upload)
 2. **Triage** — classify as invoice | PO | tax notice | irrelevant
 3. **Extract** — OCR, with AI only for low-confidence fields
 4. **Judgment rules** — trust checks
 5. If checks **pass** → Zoho push
If checks **fail** → Human review (approve / correct / reject)
 6. Later: **Zoho pull** (catch manually entered bills)
 7. Everything is written to an **audit log**
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3. How far it's built (honest status)

Layer	Status	Nuance
Data model	Done	documents, extracted_fields, judgment_results, erp_sync_log + RLS baseline
Triage	Code done	Heuristics + LLM fallback; needs live keys / inbound feed
Extract	Code done	Mindee + per-field vision LLM; needs API keys
Zoho mapping	Done	Pure rename / type convert only
Entity matching	Done	Fuzzy vendor/GL match; will not guess below confidence threshold → review
Judgment	Scaffold only	Engine exists; no real rules yet
Zoho push	Code done	OAuth + retry; blocked until judgments pass + Zoho credentials

Review UI	Done	Upload invoice, list docs, approve / correct / reject
Mailbox ingest	Not built	No Gmail / Outlook yet
Zoho pull	Not built	Still open
Team Git	Done	GitHub main + staging; feature branches → staging → main

What works today in practice

Manual upload → document appears in queue → human review decisions.
That proves the **review loop**.

What does not work hands-off yet

Gmail → auto extract → auto Zoho with no human.
That is the next build arc.

4. Important nuances (easy to misunderstand)

1. **Middleware, not ERP** — Zoho remains the system of record.
 2. **Extraction is not the product** — judgment + safe push + audit are.
 3. **Uncertainty routes to humans** — unresolved entity matches and failed rules go to review; silent “closest match” is intentionally avoided.
 4. **Local POC security is loose** — open policies for demos; must tighten before real client data.
 5. **judgment_not_passed is normal** — empty rules / no approval means Zoho push correctly refuses.
 6. **Secrets never in git** — `.env` stays local; each developer fills their own keys.
 7. **Docker / Supabase required locally** — UI and database depend on Docker engine + `supabase start`.
 8. **Staging is the team integration branch** — features merge to staging for testing, then to main.
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5. One-line status

We have a working foundation and review UI; the automated mailbox → judgment → Zoho product is designed and partly coded, but not finished end-to-end yet.

6. Sensible next build order

1. Live extract on uploaded invoices
 2. First real judgment rules
 3. Zoho sandbox push on approve
 4. One mailbox auto-ingest
 5. Auto-route: pass → push / fail → review
 6. Zoho pull + tighter auth for pilot
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Document Intelligence Connector — internal summary
Generated for co-founder / stakeholder alignment